Department of Mathematics
National Institute of Technology Kurukshetra
Statistical Distribution (MAIC 205)

Time: 3 hours | Max Marks: 50 | Roll No.: 124108055

Instruction: Calculators are not allowed. Answer each question in sequence: if you choose a question, complete all its parts together.

1. (a) Let X be a binomial random variable based on n trials and success probability p. Then mean _____ and variance _____.

   (b) Let $Y = aX + b$, then Var(Y)= _____.

   (c) A random variable X is said to have a geometric probability distribution if and only if $p(X = k) =$ _____, $k = 1, 2, 3, \ldots$ $0 \leq p \leq 1$.

   (d) A random variable X is said to have a Poisson probability distribution if and only if

$$p(X = k) = _____, \qquad k = 0, 1, 2, \ldots, \ \lambda > 0.$$

(4)

2. (a) Suppose that three fair coins are tossed, and let the random variable X denote the total number of heads obtained. Determine the probability mass function (PMF) and the cumulative distribution function (CDF) of X. Then, sketch the graph of the CDF. Also find the expectation, variance, and standard deviation of X. graph

   (b) A fair six-sided die is rolled once, and let X denote the outcome of the roll, so that $X \in \{1, 2, 3, 4, 5, 6\}$. Consider the event $A = \{2, 4, 6\}$, that is, the event that the outcome is an even number. Find the conditional probability mass function (PMF) $P(X = x \mid A)$ for all $x \in \{1, 2, 3, 4, 5, 6\}$. Then Compute the conditional expectation $E[X \mid A]$ and conditional variance $Var[X \mid A]$.

   (c) Let X be a discrete random variable with the following PMF:

$$p_X(x) = \begin{cases} \frac{1}{9}, & if x \text{ is integer in } \{-4, 4\}, \\ 0, & \text{otherwise.} \end{cases}$$

   Consider a new random variable Z, defined by the transformation g such that $Z = g(X) = |X|$. Find the PMF of Z, and then determine the expectation and the variance of Z. (4+4+4)

3. The joint probability mass function of two discrete random variables X and Y is given by

$$p_{XY}(x, y) = k(x + 2y),$$

where x and y take integer values $0 \leq x \leq 3$, $0 \leq y \leq 2$, and $p_{XY}(x, y) = 0$ otherwise.

(a) Determine the value of the constant k.

(b) Compute $P(X \leq 2, Y = 1)$.

(c) Find the marginal PMFs of X and Y.

(d) Determine whether X and Y are independent.

(8)

4. (a) Let X be a continuous random with probbility density function given by

$$f(x) = \begin{cases} kx & 0 \leq x \leq 2, \\ 2k & 2 \leq x \leq 4, \\ -kx + 6k, & 4 \leq x \leq 6. \end{cases}$$

Find the value of k and mean value of X.

(b) Use the method of least squares to fit a straight line to the five data points $(x, y) = (-2, 0), (-1, 0), (0, 1), (1, 1), (2, 3)$ **(4+4)**

5. The joint probability density function of the random variables X and Y is given by

$$f_{XY}(x, y) = \begin{cases} 2x & 0 \leq x \leq 1, \ 0 \leq y \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Find the marginal pdf of X and Y.

(b) Find the conditional density function of Y for $\{X = x\}$.

(c) Find the expectations of X and Y.

(d) Find the expectation of XY.

(e) Find the covariance between X and Y. **(10)**

6. (a) Define the correlation coefficient and explain how it measures linear dependence between variables. Describe how correlation can be used for feature selection in a multivariate dataset (e.g., removing highly correlated or low-correlation features). Finally, write a pseudocode algorithm that selects features based on their correlation with a target variable and also removes redundant features with high inter-feature correlation.

(b) The Poisson distribution with parameter $\lambda = np$ is known to approximate the Binomial distribution Binomial(n, p) under certain conditions. State the conditions under which this approximation is valid, and write pseudocode that numerically demonstrates the approximation by simulating both distributions and comparing their probability values or sample histograms.

(4+4)